

Week 1 Practical — Gradients and Laplacians of Topography

Geophysics 3A (Potential Fields & Geothermics)

May 27, 2026

Setup

Ok, let's get started by creating a Jupyter environment that we can use to run python code.

Follow the following steps if you are using a University computer:

1. From the **Start Menu**, open start typing 'python' and open **Python Command Prompt**
2. When the prompt opens, type:
`pip install scipy numpy xarray netcdf4 matplotlib ipyml`
3. Once the packages have been installed, return to the **Start Menu** and start typing 'jupyter' and open **Jupyter Notebook**
4. Once the notebook window initializes, Go to the File menu, select open from path, and open the file with the '*.ipynb' file extension.

Follow the following steps if you are using VS Code:

1. Download the codes and data file into a directory for week 1 activities.
2. Open **VS Code** and 'Open Folder' with your week 1 activities.
3. Type 'Ctrl+P' to open the command palette. Type '>' to start a command and start typing 'Python: Create Environment'. The command should appear before you finish typing so you can select it.
4. Select venv as your preferred virtual environment. You should see .venv show up as a directory in your folder.
5. Open a terminal (Ctrl+J to open the panel if needed) and type
`source .venv/bin/activate`
6. To install the required packages into the virtual environment, type
`pip install scipy numpy xarray netcdf4 matplotlib jupyterlab ipyml`

Overview

The objective of this problem set is to gain familiarity with the python environment, the use of Jupyter notebooks, and explore the use of a laplacian for smoothing an image. Implement the **2-D five-point Laplacian** for *iterative smoothing* by

$$H^{(k+1)} = H^{(k)} - \alpha \nabla^2 H^{(k)},$$

then compare how sharp features blur as smoothing increases. This builds intuition that as observations are made by **moving farther from a source produces a smoother map**.

Data

Use an **ETOPO2** NetCDF (e.g., variable z). For speed, subset a regional window (e.g., 120–160°E, 45–5°S).

Tasks

There are two major tasks as part of this problem set. The first is to complete the gradient worksheet as part of the workshop. The second will explore gradients and Laplacians of a topographic dataset in python using the file `elevation_problems.ipynb`.

Interpretation (200–300 words): Explain how curvature (large $|\nabla^2 h|$) relates to features that blur first; argue why increased smoothing mimics being farther from sources; explain why the inverse (“downward continuation”) is unstable with noise.

Extension (optional)

On a latitude–longitude grid, east–west spacing shrinks with latitude. For uniform angular steps $\Delta\phi$ (latitude) and $\Delta\lambda$ (longitude),

$$\Delta y \approx R \Delta\phi, \quad \Delta x(\phi) \approx R \cos\phi \Delta\lambda,$$

so a physically weighted discrete Laplacian is

$$\nabla^2 h \approx \frac{h_{i+1,j} - 2h_{i,j} + h_{i-1,j}}{\Delta y^2} + \frac{h_{i,j+1} - 2h_{i,j} + h_{i,j-1}}{\Delta x(\phi_i)^2}.$$

Implement this variant with per-row $\Delta x(\phi_i)$ and scalar Δy , and re-run smoothing with a smaller α .

Submission

- Figure panel: original + two smoothed maps (e.g., $K = 10, 40$), same color limits.
- Profile plot: original vs smoothed, annotated 10–90% edge widths.
- 200–300 word interpretation.
- Code (notebook/script) with your explicit Laplacian implementation.

Marking rubric (10 points)

- Correct explicit Laplacian & sensible boundary handling (3)
- Smoothing runs & plots with consistent color scales (3)
- Interpretation: curvature, distance-as-smoothing, downward continuation caution, edge-width usage (4)

Notes

Use Neumann (replicate) edges to avoid artificial sinks/sources at borders. If you implement the weighted Laplacian, choose a smaller α due to smaller Δx at higher latitudes.